Can Temporal Distance Maps Communicate Variability? A User Study with Maps of Transit Travel Times

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github.com/KentoNishi/cs271-variance

Abstract

Maps are a versatile tool for visualizing any physical area in two dimensions. With most maps, geographic distances are roughly preserved—however, geographic maps fail to depict *temporal distance*, i.e. the time it takes for a person to travel between two locations. To render travel time information, several techniques such as isocontour maps, transformed maps, and interactive radial maps are commonly used. While these existing methods successfully create an additional channel or visual encoding for travel times, they fail to communicate variability caused by factors such as traffic and transit frequency. In this paper, we propose and evaluate the efficacy of several simple extensions to existing temporal mapping methods which can simultaneously convey mean travel times and travel time variability. Our procedure is as follows: we **(1)** define "isochrone," "isoerine," "radial," and "morph" maps, **(2)** generate examples of each using real-world transit travel time data, **(3)** construct a user study to evaluate the usability and inductive biases of each variability visualization method, and **(4)** analyze the results of the user study to glean new insights. The results of our user study are highly informative for future works in the field of temporal mapping. Some of our key findings include confirmation that variance information can influence users' perceptions of relative distances, the discovery of statistically significant differences between isochrone and isoerine maps, and the characterization of scenarios in which the inclusion of variance information leads to increased confusion among users. Based on these results, we suggest several key points that designers should take into account when attempting to integrate representations of travel time variability into their maps and visualizations.

1 Introduction

Public transit maps have traditionally emphasized the physical dimensions of space, primarily focusing on distances measured in units of length. However, such representations often fail to adequately capture the practical realities of navigating urban environments, where temporal distance—defined by the time required to travel between two points—plays a significant role in the daily lives of city dwellers. Users depend heavily upon maps, often more than their own previous experiences, to craft a travel path; therefore, careful design of travel maps is critical to convey the most accurate and intuitive information to users [7].

The integration of digital mapping technologies and geospatial analytics has opened avenues for innovative approaches to cartography, enabling the visualization of urban spaces through the lens of temporal distance. Techniques such as isocontour, radial, and morphed maps have emerged to bridge the gap between physical and temporal representations of space [9, 11, 15]. Despite these advancements, existing techniques often compromise on either the accuracy of geographic details or the precision and comprehensiveness of temporal data. While conventional maps effectively depict spatial relationships, they inadequately capture the dynamic and often variable nature of mobility in urban areas, especially those reliant on public transit systems. Importantly, most maps fail to convey the *variability* and uncertainty that is inherent in travel times via public transportation—variations which can significantly affect the perceived and actual accessibility of urban areas. Previous research has highlighted the limitations of conventional transit maps in adequately representing temporal dynamics and variability [1, 10]. While these maps excel at depicting spatial relationships, they fall short in capturing the nuanced temporal aspects of urban mobility, such as fluctuations in travel times due to factors like traffic congestion, transit schedules, and service interruptions [8, 13]. Neglecting to convey this variability can lead to suboptimal decision-making and increased frustration among users.

In light of these challenges, this paper aims to advance the field of transit mapping by introducing novel methods for integrating variability and uncertainty into existing representations. By augmenting common visualization techniques with a more nuanced understanding of temporal dynamics, we seek to enhance the accuracy and usability of transit maps for urban navigation. Through rigorous user evaluations, we aim to assess the effectiveness of these methods in supporting users' decision-making processes and improving overall navigation experiences in urban environments.

This paper is composed of several contributions. First, we introduce novel approaches to integrating the variability of temporal distance into existing representations for depicting both geographic and temporal information. Through our proposed extensions of existing methodologies, we augment common techniques with a more nuanced view of urban accessibility. Then, we apply the new visualization methods to render maps based on transit travel times and variability, incorporating factors such as traffic conditions and route frequency collected from Harvard University's shuttle bus network. Finally, we conduct a user study in which we evaluate differences in users' takeaways from each map through a series of tasks and decision questionnaires based on our generated maps. With user evaluation results across 14 total tasks, we observe major differences in user decisions and preferences across combinations of advantages and disadvantages in physical distance, temporal distance, and temporal variability. Our analyses provide novel insights into the impact of 2D representations for temporal variance on users' perceptions of maps, as well as clues as to which methods best promote users to make objectively more "optimal" travel decisions.

In sum, we introduce novel graphic approaches for rendering temporal distance and variance, alongside rigorous user evaluations. In doing so, we gain a new understanding of how users interact with geospatial data. The findings of this study may be integral for enhancing the usability of such mappings and steering future methods for visualizing travel times toward more user-friendly designs.

2 RELATED WORKS

In this section, we survey various cartography-based visual encodings with time as a primary encoded element. Since our focus is on communicating transit time variability, we also highlight previous work dealing with temporal variability in visualizations.

2.1 Map-Based Encodings

Traditionally, visualizations of temporal distance come in several forms. One type is the *isochrone map*, in which temporal information is conveyed via an overlay applied to a geographically accurate map. In isochrone maps for public transportation, areas accessible within a certain time threshold from an origin point are typically represented by color or contour lines [11]. Color hue and intensity are commonly used to encode "rings" of time. There are numerous interactive tools for generating isochrone maps. For instance, CommuteTimeMap [1] allows users to dynamically move the origin point and select a time threshold in a web-based UI, while others have developed a desktop GIS application to automatically generate isochrones for public transportation maps [11]. Some platforms also provide access to the underlying data behind interactive isochrone maps, with MapBox [13] being one such platform. Isochrone visualizations successfully preserve physical distance while effectively depicting temporal distance with an extra channel, but extracting precise numbers from color gradients may be challenging.

Another type of temporal distance map is the *morph map*, in which geographic maps are warped to better represent the temporal distance between any two locations. For instance, Scheuermann et al. [15] rearranged and deformed a map of 96 cities in Germany using travel time data for train-based transportation. Nacar et al. [10] proposed a general approach for generating morphed temporal maps for any city. Transformed maps are useful for immediate interpretation since the human vision system is naturally well-suited for comparing lengths [2, 12]. However, this approach is often not scalable, as morphed maps cannot simultaneously convey geographic location or distance with temporal distance. In cities with large differences in travel time between highly accessible and highly inaccessible locations, the transformations become too severe to easily interpret relationships between locations of interest [12].

Some alternative approaches remove depictions of physical location or distance to an even greater degree, focusing entirely on temporal distance. For example, the MapBox Time Map [9] encodes time as the distance from an origin point and direction as the angle on the *radial map*. This type of map projects a location into a space where it is represented by a radius and an angle, where the radius corresponds to the distance from some point of origin and the angle corresponds to the angle from the point of origin to the point of interest as viewed on a 2D cartographic projection. Similarly, some cartographers have designed radial time-scale maps for transit networks such as the Boston T subway system [3]. While these maps can be more intuitive for representing temporal distance for well-known locations, extracting information about unlabeled locations becomes a major challenge.

2.2 Temporal Uncertainty

The vast majority of maps and map-based applications with a broad user base do not explicitly communicate uncertainty in temporal distance between two locations to the user [4, 5]. Rather, users can plan routes, view the immediate temporal distance between two points, and manually search for the temporal distance between two points at other specific times. At best, a limited set of map-based applications include a visual representation of the distribution of expected temporal distances across the span of a day. Uncertainty in a specific route primarily includes the variability in temporal distance across days and times (due to traffic and transit availability) and the reliability of transportation methods used in the route.

In terms of time variability, past works have explored methods for accurately and succinctly expressing variability in visualization. Shrestha et al. [16] propose an area-based visual encoding to convey the spatial and temporal uncertainty around events. Kay et al. [8] propose a distribution-based encoding for visualizing and modeling the uncertainty of bus arrival times in public transportation systems. These works raise the important question of visualization usability and accessibility, as complex visual encodings or statistical distributions showing variability can be difficult for many audiences to accurately interpret.

Several previous works have examined the role that visualization methods play in communicating uncertainty in data to viewers. Sanyal et al. present a framework for understanding uncertainty and test the strength of four different visual encodings via user studies [14]. The proposed encodings are size, color-mapping, glyphs, and traditional error bars, and are applied generally to both 1D and 2D data. Gschwandtner et al. [6] differentiate between statistical and bounded uncertainty by designing and testing three distinct encodings for each category. Ambiguation with color values and traditional error bars perform strongly in their respective areas; this directly informs our encoding choices on cartographic maps. Our study builds off of the visual and psychological insights generated from studies, such as the two detailed above, by grounding the uncertainty problem in the specific context of maps and travel time.

3 METHODS

In our study, we developed and tested various visualizations to effectively communicate temporal variability in public transit times. We aimed to explore how different map encodings influence user decisions when planning routes in an unfamiliar city.

3.1 Data Collection and Processing

We utilized the data from PassioGo, which encompasses vehicle times, routes, and stops from Harvard's shuttle system in Cambridge and Allston. We constructed a directed graph with stops as nodes and route segments as edges based on shuttle GPS data collected over the span of several days. With the graph, we calculated the shortest paths and the corresponding estimated walking times to/from shuttle stops. By observing the actual stop times of the shuttles over a few days, we also calculated the travel time variance between shuttle stops. To generate maps with this data, we compiled the mean time and variance results into a dense grid of 99 by 99 cells for pairwise analysis.

3.2 Map Implementations

3.2.1 Time Color Contours (Isochrone Map)

For both of the following isocontour maps, we employed a similar combination of statistical analysis and visualization tools. The map visualization starts by loading and processing the mean travel times and variance data into NumPy arrays for further operations. The variance data is processed to compute the standard deviations, which are essential for generating confidence intervals for the travel times. The coordinate system for the map is established using pre-defined geographical boundaries (bottom-left and top-right coordinates), ensuring that the data aligns accurately over the geographical area of interest.

A grid is created using linear spacing for longitude and latitude, based on the dimensions of the mean data matrix. This grid serves as the foundation for plotting the data. To create the contours, we used Matplotlib's 'contourf' to make layers of color representing different travel times at a specific confidence interval. The confidence intervals are calculated dynamically, allowing the depiction of statistical variance across the geographic grid. This is achieved by adjusting the means with a Z-score derived from the desired percentile of the normal distribution.

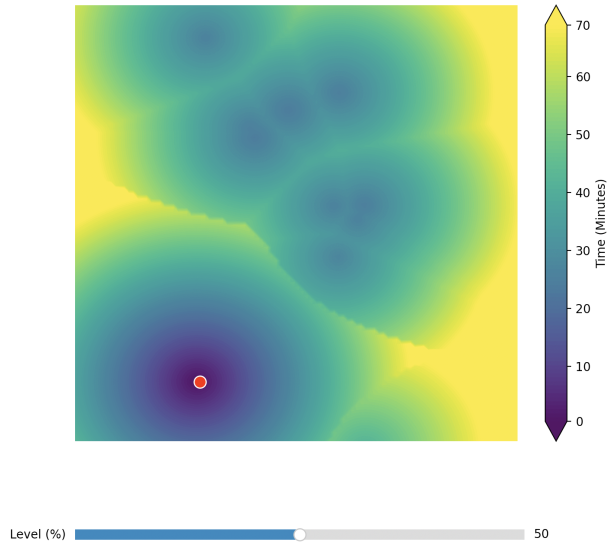

Figure 1: The time to reach places from the origin is encoded as color. The confidence level is adjusted using the slider below the map.

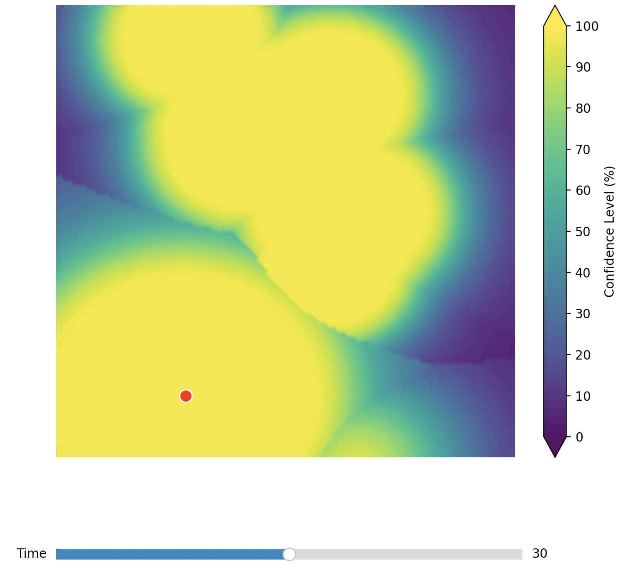

Figure 2: The confidence to reach places from the origin in the specified time is encoded as color. The specified time is adjusted using the slider below the map.

Points of interest, like the origin and significant locations, are highlighted with distinct colors and text annotations. These annotations are designed with path effects to ensure clarity against varying backgrounds. The graphical user interface elements are minimal, with the focus on the map itself. Axes labels and spines are intentionally hidden to emphasize the data visualization without the distraction of standard graph components.

An interactive slider is implemented using Matplotlib's 'Slider' widget, enabling the user to adjust the confidence level percentile. This dynamic interaction influences the travel time intervals displayed, providing a versatile tool for detailed analysis. This is optional and left out of the static images we used for user testing. An example is shown in Fig. 1.

In the static images used in our study, the contour map is overlaid with additional geographic data using Photoshop combined with custom scripting in Python's PIL module.

3.2.2 Variance Color Contours (Isoerine Map)

The implementation of the isoerine map differs from the isochrone map by focusing on visualizing varying confidence levels across the grid. While both use similar data loading and transformation techniques on the dataset, the isoerine map calculates and visualizes confidence intervals for a range of percentiles using Z-scores derived from the normal distribution. It maps these confidence levels to their geographic coordinates, creating a contour map where each contouring color indicates a different confidence level one should have when attempting to get to a location in the specified amount of time. An example is shown in Fig. 2.

3.2.3 Radial Map with Boxplot

The objective of a radial map is to abstract the angle of the location and travel time mean and variance into a polar graph, mapping our start point to the origin and the endpoint as the center of a boxplot. (The center's distance from the origin denoting mean travel time, and the boxplot's other features stemming from travel time variance).

To model this, we used a Matplotlib-based polar mapping system - starting by interpolating among mean and variance matrices to take travel time and variance from GPS coordinates, we assemble each boxplot based on these factors of angle, mean distance, and variance. We then assign the minimum, Q1, median, Q3, and maximum of the boxplot by assuming that travel times will then be Normally distributed with our aforementioned mean and variance as μ, σ^2 (in line with the Central Limit Theorem). We then assign our min and max to ± 2 standard deviations of the mean.

We can assemble the boxplot itself with a combination of radii along our pre-specified angle θ, as specified by the parameters of the boxplot. We can then draw perpendicular lines, having endpoints specified by a new radius and angle based on our desired displacement (Fig. 4). Using these lines, we can assemble the full shape of the boxplot, scaling to any arbitrary parameters for minimum, Q1, median, Q3, and maximum.

We used this code to create the maps depicting relationships between the distance, mean, and variance values, turning the set of two GPS locations into comparative boxplots similar to the one shown above. In the future, we intend to polish the code and publish it as an open-source Python library. An example is shown in Fig. 3.

3.2.4 Morph Map with Variance Color Contours

A morph map deforms distances similarly to the radial map. The primary difference is that in addition to transforming specific points of interest such that their distance from an origin point is proportional to their temporal distance from the origin, we also deform an underlying image (typically a map). Given some mesh of points within the region of interest, each mesh point is transformed; this transformation does not generate an explicit smooth or continuous function, so we use bilinear interpolation between mesh points to transform the underlying image. Morph maps less precisely communicate real-world information to users than radial maps, but increase interpretability due to the presence of an altered cartographic map.

The mesh point transformation follows from the radial map transformation. For some point p in the mesh M, defined by $p = (r, \theta)$ where r is the distance of p from the origin point and θ is the angle to p from the horizontal (defined across the origin point). Additionally, we associate some travel time t with each mesh point p, which corresponds to the temporal distance from the origin to the point p. We translate the problem such that the origin point of travel is the origin point of a Cartesian coordinate plane. We can then represent the transformation simply as follows:

$$p_{\text{original}} = (r, \theta), \tag{1}$$

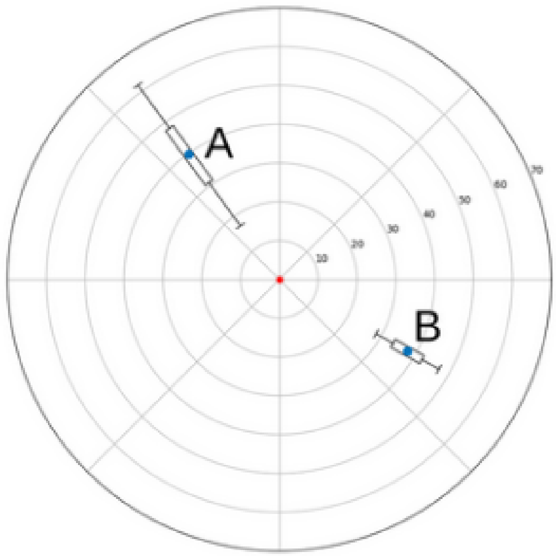

Figure 3: Radial Map Denoting `{dist:"A=B", mean:"A=B", var:"A>B"}`

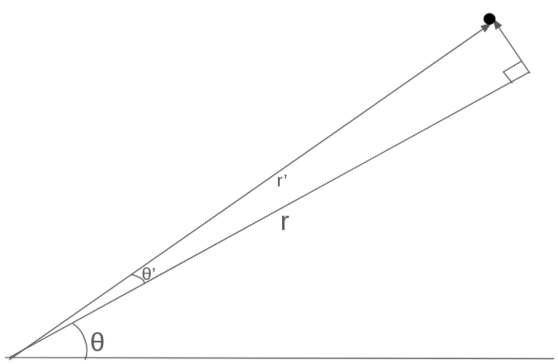

Figure 4: Angle Calculation for Perpendicular Lines in Radial Map

$$p_{\text{morphed}} = (kt, \theta). \tag{2}$$

This is true for all $p \in M$, where k is a constant determined by the size of the image and $\max_{p \in M} t$ (longest travel time to a mesh point). Note that r remains a relevant parameter for two reasons, although it is not explicitly represented in the ultimate transformation. First, morph maps are especially useful in a comparative context with an untransformed cartographic map [15]. Thus, r is explicitly represented in the original image (the object of comparison), and assists interpretability of the morph map. Second, the parameter r is used in the interpolation method to transform the underlying image, as image pixels between p_1 and p_2, each with r_1 and r_2, should remain between those two points.

The original method defined by Nacar et. al [10] takes a rectangular geographic region defined by latitudinal and longitudinal bounds, and then constructs a uniform rectangular mesh of points within those bounds [10]. Our method allows for input of an existing, not necessarily uniform mesh; without mesh input, the method constructs a uniform mesh within the provided bounds. Additionally, we extend the method beyond dependence on the Bing Maps API and allow for the input of travel data collected from other sources

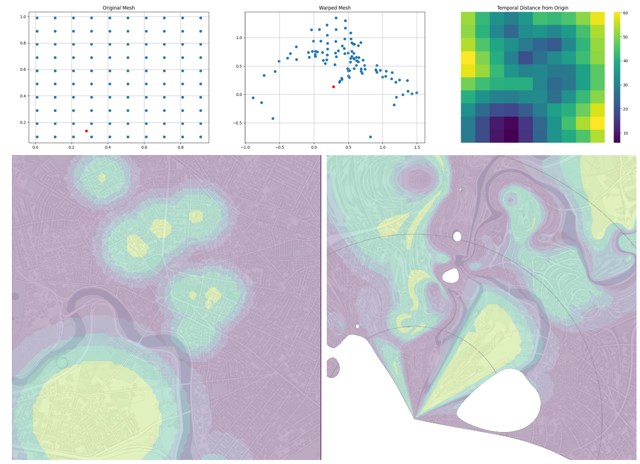

Figure 5: Mesh and image deformation in morph map. Underlying image is isoerine map controlled at 15 minutes.

(e.g. the PassioGo dataset). To include variance information in the morph map, we have two natural options given our previous methods: isoerine variance contours or boxplots at specific locations. As the morph map is best understood in the context of an untransformed map, we apply isoerine variance color contours to the underlying cartographic map, which then becomes deformed according to the mesh. An example is shown in Fig. 5.

Unfortunately, the method performs quite poorly on the PassioGo dataset specifically. Our observed region is quite small, and is split into relatively fast shuttle transport and relatively slow walking; these features induce sharpness into our data landscape. This sharpness is the enemy of the morph map, as points that are further away in space become very often closer in time, especially for small angular ranges along the shuttle path. Holes appear on the map due to discontinuities in travel time, as conic regions radiating from shuttle stops along the shuttle path are pulled toward the origin. This was partially resolved by lowering the mesh resolution, thereby increasing geographic space between mesh points and smoothing interpolated values; however, it did not fully resolve the issue. This effect is smoothed for large public transportation networks in cities. Further investigation into the datasets and geographic scales in which the morph map is interpretable and useful is necessary.

4 User Study Design

We aimed to evaluate user preferences and decision-making processes when exposed to various map types, focusing on the influence of visual encodings of travel time and its variability on user choices between two potential destinations. Participants were recruited via Prolific, based in the US, and each survey involved 35 participants. The estimated completion time for the survey was 5 minutes, and a total budget of $50 was allocated for each survey session. We utilized Google Forms for its ease of use to collect responses and prompted participants to reason behind their choices to mitigate the risk of random inputs.

Participants were asked to choose between two locations, A and B, under different combinations of physical distance, mean travel time, and travel time variability. An example is shown in Fig. 6. Each map type was tested under controlled conditions to isolate the effects of each visual encoding on user decisions. To compare user preferences across mapping methods, we applied the chi-squared statistical significance test of homogeneity—this allows for the identification of significant differences in user preferences across different map types under the various conditions set in the experiment.

Due to time and cost constraints, we ran our experiments for the

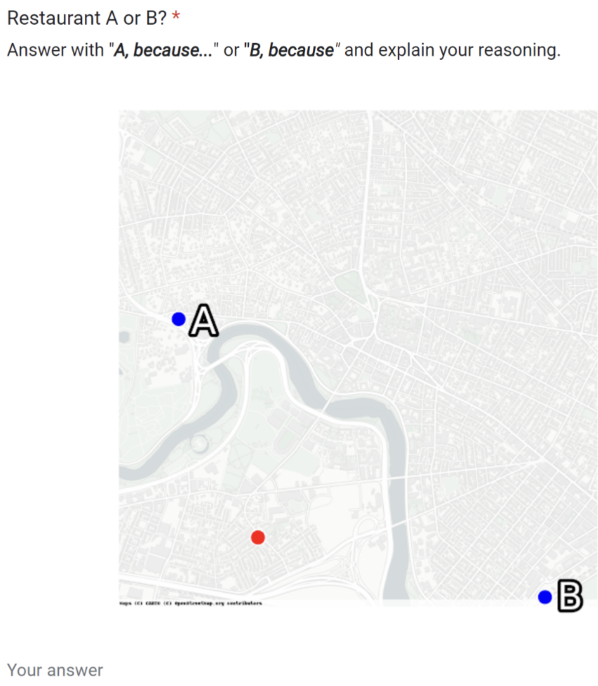

Figure 6: A sample question included in the baseline survey.

baseline, isochrone, and isoerine methods only. We leave complete user studies with the radial and morph map methods for future work.

5 RESULTS

We present the full results of our user study in Tab. 1. Some of our most interesting results, labeled "Key Result," have intuitive interpretations that hint toward the characteristics of each mapping method we tested. In this section, we detail our understanding of each of the Key Results.

KR1: We note that the {`dist:"A=B", mean:"A=B", var:"A<B"`} scenario (where our physical distances and mean travel times are equal, and the only differentiator is variance), sees our baseline rendering have a 14-to-21 split favoring location B, while our isoerine and isochrone renderings both have a slightly stronger 10-to-25 split favoring location B.

We find that the proportion of isochrone and isoerine preferences differs slightly from the baseline, but not enough to be statistically significant - their rendering of increased variance was not enough to nontrivially sway the user towards the location with less variance. If anything, the effect seemed to be the opposite, with users slightly favoring location B. It follows that variance by itself had a lack of impact on user choice.

In a more abstract sense, this signals that the variance alone has a negligible effect on the user when distance and travel time seem similar. Within this context, there is not much perceived difference between routes of differing variance.

KR2: When both the mean travel time and variance are in favor of one location over another ({`dist:"A=B", mean:"A<B", var:"A<B"`}, {`dist:"A=B", mean:"A>B", var:"A>B"`}), users reliably ignore physical distance and choose the travel time-optimizing option.

On a cartographic map with no additional information, participants slightly favored (20:15) the temporally closer option. With a small sample size, it is difficult to argue whether participants intuited a temporally shorter path to one location, or if this was largely a random choice. The isochrone representation of this problem resulted in participants heavily favoring the temporally closer location (29:6); the isoerine map resulted in similarly strong preference (32:3).

This is a simple baseline that meets expectation. From this, we claim that our representations of time and variance are generally interpretable and useful in the simplest case decision-making scenario. Additionally, this suggests that our colormap chosen for the visualization is interpretable for both isoerine and isochrone maps, where warm and cool colors represent opposite degrees of preferability across the mapping techniques. This does not mean at all that we necessarily follow best practices at every step, but that the existing visualization is useful to an audience unfamiliar with vis and map transformation techniques.

KR3: This scenario tested user behavior when A is geographically closer than B with equal travel times, A has less variability in arriving in the expected time (`dist:"A<B", mean:"A=B", var:"A<B"`}). A is strongly preferred in both the baseline and isochrone settings (33:2), and there is no difference between these 2 representations, matching expectation.

However, the isoerine map decreases the preference for A (26:9). This could be due to several reasons. Some users may be confusing the color scales in isochrone and isoerine maps, some may be confused by the visual inclusion of variance, and some especially risk-seeking users may prefer the travel time with higher variability. In the color contour maps using the Viridis colormap, yellow regions correspond to high values, while blue regions correspond to low values. In isochrone maps, we prefer regions with low values (low expected time of arrival); however, in isoerine maps, we generally prefer regions with high values (high confidence in arriving in the expected time). The consistency of color coding in our design aligns with numerical value, rather than with user preference, which is something to consider for further design iterations.

KR4: This scenario tests when A is geographically closer than B, A is temporally further than B, and they have equal variance in travel times (`dist:"A<B", mean:"A>B", var:"A=B"`}). Assuming that users are optimizing to minimize travel time, B should be preferable. In the baseline map, all users prefer A (35:0), as it is geographically closer. The isoerine map fails to persuade users that B is preferable; this supports the hypothesis that the isoerine map is either uninterpretable (due to the complexity of travel time variance intuition) or actively misleading (due to color scale).

The isochrone map only slightly pushes users to prefer B (29:6), and this is statistically significant. However, the effect is weaker than desired. This suggests that isochrone and isoerine maps do not work particularly well with equal variance but contradict physical distance and mean travel time.

Summary: Most experiments meet expectation to some degree when one destination is clearly preferable to another. However, when intuition is the opposite for geographic distance and temporal distance, users tend to make decisions based on the map with which they are more familiar (baseline cartographic map). The isoerine map (variance color contours) is especially confusing, likely due to both the color encoding and the difficulty for users to interpret variability in travel time. Isochrone maps are generally at least somewhat effective in communicating the optimal destination, and often strongly effective. These maps are best in the decision-making space when they align with the baseline intuition.

Distance	Mean	Variance	Method 1	A	B	Method 2	A	B	PValue	Key Result
$A=B$	$A=B$	$A=B$	baseline	17	18	isoerine	18	17	1.000	
$A=B$	$A=B$	$A=B$	baseline	17	18	isochrone	17	18	1.000	
$A=B$	$A=B$	$A=B$	isoerine	18	17	isochrone	17	18	1.000	
$A=B$	$A=B$	$A<B$	baseline	14	21	isoerine	10	25	0.450	
$A=B$	$A=B$	$A<B$	baseline	14	21	isochrone	10	25	0.450	KR1
$A=B$	$A=B$	$A<B$	isoerine	10	25	isochrone	10	25	1.000	
$A=B$	$A<B$	$A=B$	baseline	19	16	isoerine	31	4	**0.003**	
$A=B$	$A<B$	$A=B$	baseline	19	16	isochrone	33	2	**0.000**	
$A=B$	$A<B$	$A=B$	isoerine	31	4	isochrone	33	2	0.673	
$A=B$	$A<B$	$A>B$	baseline	27	8	isoerine	29	6	0.766	
$A=B$	$A<B$	$A>B$	baseline	27	8	isochrone	31	4	0.342	
$A=B$	$A<B$	$A>B$	isoerine	29	6	isochrone	31	4	0.734	
$A=B$	$A<B$	$A<B$	baseline	20	15	isoerine	29	6	**0.036**	
$A=B$	$A<B$	$A<B$	baseline	20	15	isochrone	32	3	**0.002**	KR2-A
$A=B$	$A<B$	$A<B$	isoerine	29	6	isochrone	32	3	0.477	
$A<B$	$A=B$	$A=B$	baseline	35	0	isoerine	27	8	**0.005**	
$A<B$	$A=B$	$A=B$	baseline	35	0	isochrone	24	11	**0.000**	
$A<B$	$A=B$	$A=B$	isoerine	27	8	isochrone	24	11	0.592	
$A<B$	$A=B$	$A>B$	baseline	27	8	isoerine	29	6	0.766	
$A<B$	$A=B$	$A>B$	baseline	27	8	isochrone	31	4	0.342	
$A<B$	$A=B$	$A>B$	isoerine	29	6	isochrone	31	4	0.734	
$A<B$	$A=B$	$A<B$	baseline	33	2	isoerine	26	9	**0.045**	
$A<B$	$A=B$	$A<B$	baseline	33	2	isochrone	33	2	1.000	KR3
$A<B$	$A=B$	$A<B$	isoerine	26	9	isochrone	33	2	**0.045**	
$A<B$	$A>B$	$A=B$	baseline	35	0	isoerine	35	0	1.000	
$A<B$	$A>B$	$A=B$	baseline	35	0	isochrone	29	6	**0.025**	KR4
$A<B$	$A>B$	$A=B$	isoerine	35	0	isochrone	29	6	**0.025**	
$A<B$	$A>B$	$A>B$	baseline	33	2	isoerine	9	26	**0.000**	
$A<B$	$A>B$	$A>B$	baseline	33	2	isochrone	5	30	**0.000**	KR2-B
$A<B$	$A>B$	$A>B$	isoerine	9	26	isochrone	5	30	0.371	
$A<B$	$A>B$	$A<B$	baseline	32	3	isoerine	25	10	0.062	
$A<B$	$A>B$	$A<B$	baseline	32	3	isochrone	19	16	**0.001**	
$A<B$	$A>B$	$A<B$	isoerine	25	10	isochrone	19	16	0.216	
$A<B$	$A<B$	$A=B$	baseline	27	8	isoerine	33	2	0.084	
$A<B$	$A<B$	$A=B$	baseline	27	8	isochrone	34	1	**0.028**	
$A<B$	$A<B$	$A=B$	isoerine	33	2	isochrone	34	1	1.000	
$A<B$	$A<B$	$A>B$	baseline	24	11	isoerine	29	6	0.265	
$A<B$	$A<B$	$A>B$	baseline	24	11	isochrone	30	5	0.153	
$A<B$	$A<B$	$A>B$	isoerine	29	6	isochrone	30	5	1.000	
$A<B$	$A<B$	$A<B$	baseline	33	2	isoerine	35	0	0.493	
$A<B$	$A<B$	$A<B$	baseline	33	2	isochrone	35	0	0.493	
$A<B$	$A<B$	$A<B$	isoerine	35	0	isochrone	35	0	1.000	

Table 1: Comparison of baseline, isochrone, and isoerine mapping methods across all combinations of distance, mean, and variance relations. P values are obtained via the chi-square test for homogeneity. Key results are labeled in the rightmost column.

6 Discussion

In this study, we explored how temporal distance maps can be utilized to effectively communicate the variability in transit travel times, a factor that is a critical but often overlooked part of public transportation tools. Our study provides insights into the design and efficacy of different visual encodings in conveying various relationships between physical distance, temporal distance, and temporal variability.

6.1 Integrating Variability Into Temporal Maps

Temporal variability is an element of travel times that is often neglected within maps. Our implementation of three visual encodings of temporal variability—time color contours (isochrone map), variance color contours (isoerine map), and a radial map with boxplots—demonstrate the diversity of approaches that can be used to convey variability. In addition, in implementing these different encodings, we recognize that viewability vs. information density is an important tradeoff that must be especially considered when incorporating variability into temporal maps, as variability can be a complex statistical concept that many lay users may struggle to conceptualize at first glance of a visual map. This tradeoff is reflected in the results of our user study, where users struggled to incorporate variability into their mental model when temporal travel times were similar.

6.2 User Behaviors

Our user study provides rich insight into the interaction between users and various visual encodings of temporal variability in travel maps. In particular, we examine how different map types with encodings of variability (baseline, isoerine, isochrone) affect user choices in a simulated decision-making context involving urban navigation. Overall, our key results highlight that participants responded differently to the visual encodings, suggesting that map design can significantly influence decision-making in urban navigation.

Key Result 1 (KR1) showed that variability information alone (i.e., when mean travel times are equal but variability differs) has little effect on user preference. This suggests that users might not inherently value or even notice variability unless it is coupled with significant differences in mean travel times or communicated in a more impactful way, challenging the assumption that simply presenting additional data (like variability) would automatically lead to better decision-making. Overall, this underscores the need for map designs to not only present data but also contextualize it in a way that aligns with users' decision-making priorities.

Key Result 2 (KR2) and **Key Result 3 (KR3)** further illustrate how users prioritize different types of information. In **KR2**, users consistently chose locations with lower mean travel times and lower variability, even if those locations were physically further away. This indicates a clear preference for reliability in travel time, suggesting that visual encodings that effectively communicate both travel time and its reliability could significantly influence user decisions. However, in **KR3**, we observe an interesting countereffect where the isoerine visualization led to confusion rather than clarity, causing users to make less optimal decisions based on the information provided. Furthermore, the isochrone (confidence as the controllable variable, time as contour) maps generally outperformed isoerine maps in helping users make more "optimal" travel decisions, particularly when variance was a critical factor in their choice.

Another intriguing finding was that even when physical and mean travel times were similar, the additional information about travel time variability did not significantly influence user choices (**KR1**). This suggests that unless travel time variability is substantial, users may not factor it into their decision-making processes as strongly as hypothesized. This could be due to the cognitive load of processing additional data layers or a lack of user familiarity with interpreting such information, challenges that could be addressed through more careful design of how variability can be encoded in temporal maps.

6.3 Design Implications for Temporal Maps

The observations from our study have several implications for the design of maps that simultaneously visualize travel time, physical distance, and variability. First, our key results demonstrate that is evident that conveying temporal distance can help users overlook biases within physical distance. Since most maps currently only show physical distance, incorporating more measures of temporal distance (such as through isochrone color gradients) can help enhance the accuracy and intuition of visual maps.

Furthermore, Our findings indicate that in the context of real-time decision-making, such as selecting transit routes, the importance of simplicity and clarity in conveying variability cannot be overstated. Notably, our key results demonstrate the challenge users face in incorporating variability into their decision-making processes concerning travel routes. Complex visualizations tend to exacerbate this challenge, potentially leading to user confusion rather than facilitation. Thus, designers should carefully consider user expertise and the situational context during the development of such tools. Tailoring the complexity and interactivity levels of these tools based on user preferences or skill levels could prove beneficial in enhancing user comprehension and decision-making efficacy.

7 Future Work

Enhanced Communication of Variability: Undoubtedly, there are more visual encodings that can be used to integrate variability into temporal maps, and this study presents an investigation into a few types that can be further explored upon. Since variance alone does not significantly impact decision-making unless highlighted effectively, future designs should explore more intuitive or prominent methods of visualizing variability—such as dynamic elements or interactive features that allow users to explore how variability affects travel time under different conditions.

User-Centric Design Approaches: The challenge remains to design these maps in a way that they are not only informative but also intuitive and aligned with the cognitive processes of users. Map designs should be developed with a deeper understanding of user preferences and cognitive processes. This involves not only the visual representation of data but also the interaction design—how users interact with the map and access the information. Future work should focus on refining visual encodings, expanding user testing to more diverse scenarios, and integrating real-time data to increase the accessibility, responsiveness, and utility of these maps in everyday urban navigation.

Broadened User Studies: We present and implement four visual encodings of variability in this work; however, as we were limited due to budget constraints, we were only able to run a user study on the following setups: a control baseline, the isoerine visualization, and the isochrone visualization. With more resources, the same user study could be run on the other map types (morph map, radial map) that we implemented in addition to the baseline and isochrone map. Future work should include a wider variety of map types and a larger, more diverse participant pool to validate and refine the findings. This could include more interactive elements in user studies to see how engagement with the map affects understanding and decision-making.

8 Conclusion

This study presents a novel user evaluation of temporal distance maps, specifically focusing on how the encoding of travel time variability influences urban navigation decisions. Using Harvard University's shuttle system data, we applied standard map encodings such as isochrone and radial maps to investigate user preferences in a simulated navigation scenario. Key findings include the effectiveness of confidence-controlled isochrone maps in aiding optimal decision-making and the need for prominent and intuitive presentation of variability information. The potential of temporal distance maps to enhance urban mobility is significant, yet future designs must be user-centered and intuitive to improve real-world navigation. Future research should focus on refining visual encodings and integrating real-time data to enhance the utility of these maps in everyday urban scenarios.

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